

Testing Javascript with Mocha, Chai, Sinon



Mocha is a **JavaScript test framework** that runs your tests and reports the results.

1. Finds your test files.
2. Executes the tests.
3. Tells you which tests passed and which failed.

<https://mochajs.org/>

<https://www.chaijs.com/>

<https://sinonjs.org/>

Mocha

Mocha provides, These help you structure, setup and breakdown the tests.

- describe()
- it()
- before()
- after()
- beforeEach()
- afterEach()
- ``

```
describe('addNums', () => {  
    it('should add two numbers', () => {  
        expect(addNums(2, 3)).to.equal(5);  
    });  
});
```

Mocha does not provide assertions (e.g. expect/should) This is provided by an assertion library like chai.

Mocha, Chai, Sinon relationship

Mocha – runs tests

Chai – provides assertions

Sinon – stubbing, spying and mocking in javascript tests.

Mocha runs the test, Chai checks the results, and Sinon lets you control and inspect the dependencies around the code you're testing.

This is but one test stack that is available. Jest and vite/vitest are other possible alternatives

<https://mochajs.org/>

<https://www.chaijs.com/>

<https://sinonjs.org/>

Sinon Core Features

Spies: Record information about function calls, such as how many times they ran and what arguments they received, without changing behavior.

Stubs: Replace a function or method entirely, letting you control behavior by forcing custom return values or throwing errors.

Mocks: Combine spies and stubs with pre-programmed expectations upfront to verify specific usage patterns on an object.

Fake Timers: Control and manipulate time-dependent functions like `setTimeout` or `setInterval` during tests.

Fake Servers: Simulate and pre-program responses for AJAX and network requests.

Use Sinon when your code depends on something external:

- Database
- File system
- Email service
- REST API
- Logger
- Time/date functions
- Other modules

Don't use Sinon for simple calculations or pure functions.

Sinon (spy, stub, fake)

- Sinon is important because it makes the tests reliable, isolated and focused. Sinon helps us to isolate from dependencies within the function. Without a library like Sinon you can end up testing more than you actually care about.
- If you have a function that makes a call to a live mongo database and the test fails
 - is it because the MongoDB is down.
 - Is the network slow
 - Has authentication failed etc...

Testing in Angular

The default test stack for Angular 21/22 is:

- Vitest (Runner and assertion library and mocking/spying)
- Angular TestBed - lets you create Angular components, services, directives, and pipes in a way that closely resembles how Angular creates them at runtime
- Jsdom - creates a fake browser environment

Jasman and karma are no longer the default choice but are available to use.